

**Parameter**

Sulfide

Sample Type

Wastewater

Introduction

The ion-selective electrode method is an EPA-approved test procedure to directly read sulfide in wastewater. Laboratory productivity is improved with quick, uncomplicated determinations. Use of the SAOB frees sulfur from hydrogen ion (by buffering at a pH above 12), prevents oxidation of sulfide, ensures that samples and standards have similar ionic strength, and reduces the effect of interfering ions. This procedure conforms to EPA-approved test procedures from Standard Methods and ASTM, as listed below.

References

1. Method 4500-S²⁻ G. Standard Methods for the Examination of Water and Wastewater, 20th Edition, 1998. APHA, AWWA, & WEF, Washington, D.C. www.standardmethods.org
2. D4658-03. Annual Book of ASTM Standards, Section 11, Water and Environmental Technology, Volume 11.01, 2005. ASTM International, West Conshohocken, PA. www.astm.org

Result Statistics

# Trials	Average	% Recovery	%CV
4	0.52 mg/L*	104%	8.7 %

*Note: Sulfide was not detected in wastewater sample; standard was spiked into the samples at 0.5mg/L and the above results were obtained.

Recommended Equipment

5-Star Plus Benchtop pH/ISE/Cond/DO meter (Orion 1119000) or other Orion ISE meter; ionplus® silver/sulfide electrode (Orion 9616BNWP); benchtop stirrer (Orion 096019). Optional: printer (Orion 1010006); Star Navigator Software (Orion 1010007).

Required Solutions

Saturated Sulfide Stock, Daily sulfide calibration standards (66.5, 13.3, 1.33 and 0.665mg/L), Sulfide Anti-Oxidant Buffer (SAOB) (941609), Optimum Results A fill solution (900061), 0.1M Lead Perchlorate (948206), **Zinc Acetate (2M)**, **Sodium Hydroxide (6M)**, Deionized water

Solutions Preparation

Zinc acetate (2M): Dissolve 43.90g of zinc acetate dihydrate in water, using 100mL volumetric flask, dilute to 100mL with water.

Sodium Hydroxide (6M): Dissolve 24g NaOH in 100mL of water.

Saturated Sulfide Stock: dissolve 100g sodium sulfide nonahydrate (Na₂S•9H₂O) in 100mL DI water, mix and let sit overnight.

Intermediate Sulfide Standard: pipette 1.5mL of sulfide stock into 100mL of DI water; titrate with lead perchlorate to determine exact concentration, good for 1 week. See Page 2 for details.

Daily Sulfide Standards: dilute the intermediate std with DI water to prepare calibration standards of 0.665, 1.33, 13.3 and 66.5mg/L (or 4 standards with similar concentrations as those listed). Dilute each calibration standard 1:1 with SAOB to prevent the oxidation of sulfide and add zinc acetate sodium hydroxide to all standards if analyzing preserved or PT (proficiency testing) samples. Calibrate/verify calibration with 20mL of these standards as they have already been diluted 1:1 with SAOB. See Page 2 for details.

Meter Setup

Connect the electrode and stirrer to the Star Meter. Set measurement mode to ISE. In Setup mode, set resolution to 3, set the range to high, select mg/L as the unit, and set read type to auto. If all steps were followed correctly, the meter display will show three digits in the top line and "ISE: mg/L" to the right of the top line.

Electrode Setup

See the electrode manual for preparation of the electrode. Use Optimum Results A filling solution to obtain best results.

Electrode Performance Check

Check slope at least daily according to the electrode manual. Drift may be checked by comparing a 1-minute to a 2-minute reading. Results should agree with desired criteria. See troubleshooting section of manual for slope or drift problems.

Electrode Storage, Soaking, and Rinsing

See electrode manual for storage 1) between measurements, 2) overnight, and 3) for long periods of time. Between measurements, rinse the electrode with DI water and dry outer sleeve of electrode to remove drops before measuring the next sample. Do not wipe or rub the sensing element of the electrode.

Sample Preservation

If sample cannot be measured immediately, preserve in aliquots to be tested (for example 50 or 100mL) with zinc acetate and sodium hydroxide. 4 drops of 2M ZnOAc and 1 drop of 6M NaOH should be added to every 100mL of sample. Refer to references and/or EPA 40 CFR Part 136.3 for details.

Sample Preparation

Unpreserved sample: Measure 10mL of sample and 10mL SAOB into a 50mL beaker. Preserved sample: add SAOB in 1:1 ratio to entire aliquot of preserved sample (for example 50 or 100mL, see sample preservation) to dissolve precipitate and free sulfide. Measure 20mL aliquots of the sample+SAOB solution into 50mL beakers. For precise measurements, allow all standards and samples to reach the same temperature before analysis. **PT sample: dilute 10mL of concentrated standard into DI water, use 1L volumetric flask. Add 2mL of 2M ZnOAc and 0.5mL 6M NaOH and bring to volume with DI water.**

Calibration

Perform a four point calibration using the 4 Daily Sulfide Standards (see Solutions Preparation). After calibration, the electrode slope will be displayed and should be above 25mV/decade. Read a fresh portion of a standard to verify calibration. If reading is not acceptable, see troubleshooting section of electrode manual.

Analysis

Rinse electrode and stirrer with DI water and dry outer sleeve of electrode to remove drops. Place probe and stirrer in sample and measure. The concentration of the sample will be displayed. When a stable reading is achieved, the "ISE:mg/L" icon will stop flashing, the concentration will be displayed in mg/L and the stir probe will turn off.

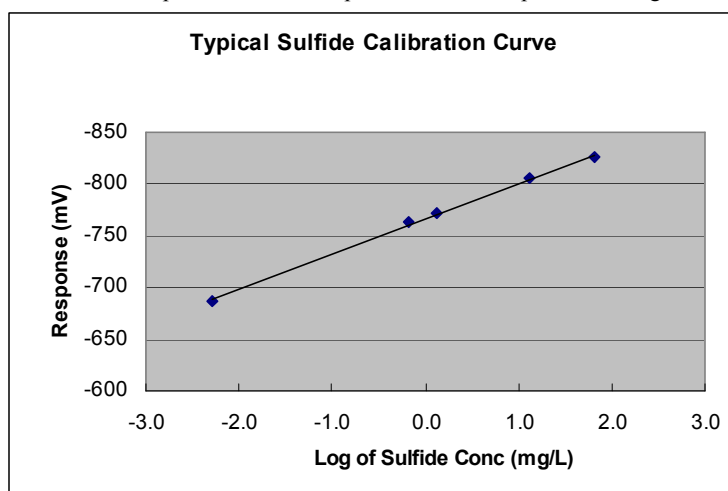
Quality Control (QC)

Recommended QC procedures include: calibration and calibration verification, initial demonstration of laboratory capability and method detection limit determination, laboratory control samples (LCS), method blanks, matrix spikes (MS), sample duplicates, and independent reference materials. See references above for details.



Wastewater*	ppm (mg/L) Sulfide
Sample 1	0.58
Sample 2	0.48
Sample 3	0.52
Sample 4	0.49
Mean	0.52
Average Recovery	104%
Standard Deviation	0.045
%CV	8.7%

*Note: Sulfide was not detected in wastewater sample: standard was spiked into the samples at 0.50mg/L and the above results were obtained.



Titration Intermediate Sulfide Standard (IS)

Measure 20mL of the Intermediate Sulfide Standard into a 50mL beaker. Titrate with 0.1M Lead Perchlorate (948206) and monitor with the Silver/Sulfide (9616BNWP) Electrode. Typically 4-6mL of titrant is required to reach endpoint. It is recommended to perform 3 titrations of the standard and average the results. Multiply the average volume of titrant required by 160 to determine the concentration of the standard in mg/L. See References for details.

Preparing Daily Sulfide Standards

Once the Intermediate Sulfide Standard concentration has been determined by titration, this solution is diluted to make up 4 Daily Sulfide Standards. **Do not add zinc acetate and sodium hydroxide if only analyzing non-preserved samples.** Prepare 500mL of the most concentrated standard (66.5mg/L) and prepare 100mL of each subsequent standard. Start by measuring 250mL of SAOB, **1.0mL of zinc acetate and 0.250mL of sodium hydroxide** in a 500mL volumetric flask. Calculate the amount of Intermediate Sulfide Standard required needed to prepare 66.5mg/L std (shown below; use 250mL volume so std are diluted 1:1 as samples will be) and add to the SAOB. Bring up to volume with DI water. Dilute the 66.5mg/L standard to prepare the remaining calibration standards as shown below.

66.5 mg/L: Vol IS required = (250mL)*(66.5mg/L)/(concentration of IS) = 16625/(concentration of IS)

For example, if IS is determined to be 720mg/L,

Vol IS required = 16625/720 = 23.1mL to 500mL flask

13.3 mg/L: Measure 10mL of 66.5mg/L std into a 100mL flask with 50mL of SAOB, **4 drops of zinc acetate and 1 drop of sodium hydroxide**, bring up to volume with DI water

1.33 mg/L: Measure 1mL of 66.5mg/L std into a 100mL flask with 50mL of SAOB, **4 drops of zinc acetate and 1 drop of sodium hydroxide** bring up to volume with DI water

0.665mg/L: Measure 0.5mL of 66.5mg/L std into a 100mL flask with 50mL of SAOB, **4 drops of zinc acetate and 1 drop of sodium hydroxide** bring up to volume with DI water